Practical Incentive-Compatible Drafts: Carry-Over Lottery Allocation

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Open-source repository: https://github.com/kvr06-ai/cola-sloan-ssac27

1. Introduction

On February 7, 2026, in a National Basketball Association (NBA) game, the Utah Jazz led the Orlando Magic 94-87 going into the fourth and final quarter. The Jazz then kept two of their best players, Lauri Markkanen and Jaren Jackson Jr., out of the game for the whole quarter. Orlando outscored them 33-23 and won 120-117 [1]. Five days later, the NBA fined the Jazz $500,000 over that game and one other [2]. The Jazz had company. The Ringer, a sports website, counted eight of the league's thirty teams trying to finish with the worst records that month [3]. Losing on purpose in this way is called tanking.

Those teams wanted a better place in the draft, the yearly event in which NBA teams take turns choosing new players. Each turn is called a pick, and earlier picks give teams a better choice. In 2026, the available players were led by AJ Dybantsa and Darryn Peterson [3]. Sixteen of the thirty teams reach the playoffs, the tournament that follows the regular season and decides the champion. The fourteen teams that miss the playoffs pick first in the draft, and a lottery, which is a random drawing, decides which of them get the earliest picks. Under the lottery the NBA used from 2019 to 2026, the three teams with the worst win-loss records each had a 14% chance at the first pick. The team with the fourteenth-worst record had a 0.5% chance. The drawing decided only the first four picks. The rest went in reverse order of record, which means the worst remaining team picked next [4]. A worse record therefore meant better odds in the drawing and an earlier pick. Taylor and Trogdon [5] compared three NBA seasons played under different draft rules and found that teams lost more often when the rules rewarded losing.

The draft exists to help weak teams, and that goal creates the problem. At this conference in 2020, Banchio and Munro [6] studied lotteries that set their odds by the end-of-season standings, the final ranking of teams by record. They proved that every such lottery gives some team a reason to lose on purpose, unless it gives all the teams that miss the playoffs the same odds. A lottery of this kind therefore cannot both favor the worst teams and remove the reason to lose. We call this the impossibility result. Every change the NBA has made to its lottery has traded one of these two goals against the other. The latest, called 3-2-1, was adopted in May 2026. It gives the three teams with the worst records fewer chances in the drawing than the seven teams just above them. The teams at the bottom therefore no longer gain by losing [7]. A team in the middle of the standings still can, because of the play-in tournament. The NBA is split into two conferences, East and West. In each one, teams that finish seventh to tenth enter the play-in, a short qualifying round for the last playoff spots. Under 3-2-1, the teams that finish ninth or tenth get more chances in the drawing than those that finish seventh or eighth. A team in that range therefore improves its chances by finishing lower [7]. The new lottery also sets its odds by the current season alone, so a team that has been weak for many years gets no extra help.

The impossibility result depends on measuring weakness by a single season's standings. We introduce Carry-Over Lottery Allocation (COLA), which measures weakness by playoff results over many seasons. Each team has a draft priority, a score that carries over from one season to the next and gives the method its name. A team's priority grows each season it goes without playoff success. It falls when the team wins a playoff series, which is one round of the playoffs between two teams, or when it receives a top pick. COLA works like a waiting list. One more loss this season leaves a team's place on the list unchanged, so a draft that follows the list need not reward losing. Help still reaches weak teams, because the teams that stay weak are the ones that wait longest.

COLA is a family of rules for setting the draft order, and we call each such rule a mechanism. Full-lottery COLA holds a drawing for every pick among teams that miss the playoffs, and a team's chances are proportional to its priority. Simple COLA has no drawing. The team that has waited longest picks first. The NBA's 3-2-1 can be read as an extreme member of the family. This season's standings alone set its priority, and it carries only one thing from year to year: a limit on how often the same team can win the top picks. The same priority could also be used in an auction, a different kind of draft in which teams bid for players or for draft positions. It would decide how much each team has to bid.

We focus on Simple COLA. For each team, it counts the seasons since the team last won a playoff series or received one of the first three draft picks. The team with the highest count picks first. If two teams have the same count, the one with more wins this season picks ahead of the other. When no team can improve its pick by losing on purpose, a mechanism is called incentive compatible. We prove that Simple COLA is incentive compatible. The proof rests on a few assumptions about how teams behave, and all but one are standard. The exception, which Section 4 defends, is that a team would rather win a game than lose it to change which other teams it competes with for the top picks.

We test nine mechanisms, including the NBA's old lottery and 3-2-1, in ZenGM Basketball-GM. It is a simulator whose code is public, and it simulates every game and every change to each team's roster in realistic fictional leagues. Each mechanism runs on the same 30 leagues for 25 seasons. We compare the mechanisms on three measures. The first is how much they help the teams that have waited the longest. The second is how evenly playoff success spreads across teams over many seasons, which we call long-term parity. The third is the reward for losing, meaning how much a team's pick improves when it loses more games. Because every mechanism plays the same 30 leagues, we can compare any two of them league by league. For each comparison, we report a 95% confidence interval, the range that very likely contains the true difference. The leagues are fictional, so we also check whether the reward for losing and the help for long-waiting teams look the same on real NBA standings, one season at a time.

Of the nine mechanisms, Simple COLA does the most for the teams that have waited longest. Under 3-2-1, the longest any team goes without a playoff series win or one of the first three picks averages 16.3 seasons. Under Simple COLA, it averages 5.5. No team that misses the playoffs ever gains by losing more games. The average wait for a team's next playoff series win is within about a tenth of a season of the wait under 3-2-1. Simple COLA has two costs, which Sections 5 and 7 measure. First, teams wait about 0.2 seasons longer to return to the playoffs. Second, a team that has waited a long time can gain by losing a first-round playoff series. A series win would send it to the back of the list. Every number in this paper can be reproduced with the code in our public repository. The 3-2-1 lottery expires after the 2029 draft unless the league votes to keep it [7], so the league will soon choose a draft mechanism again.

2. Background

Teams have tanked more in the years when the draft paid more for losing. Until 1984, the last-place teams in both conferences flipped a coin for the first pick, and the other teams picked in reverse order of record. For the next five drafts, a lottery gave every team that missed the playoffs the same chance at the first pick, a format we call equal odds. In 1990, the NBA weighted the lottery so a worse record brought a better chance [6]. Taylor and Trogdon [5] compared three seasons, one played under each format. They looked at teams that could no longer reach the playoffs, which we call eliminated teams. They found tanking by eliminated teams under reverse order and under the weighted lottery, and none under equal odds. Price et al. [8] extended the test to every game from 1977 to 2007 and found more late-season losing in the seasons when finishing last paid the most. Bookmakers set their odds as if eliminated teams tank, although the same study finds mixed evidence of tanking in the actual scores [9]. Fornwagner [10] finds that National Hockey League teams also lose more after elimination.

Banchio and Munro's impossibility result explains why equal odds were the exception, and its proof needs only two teams [6]. Take two eliminated teams with the same record that meet in the last game of the season. If finishing lower gives a better chance at the first pick, the losing team comes out ahead, so neither team wants to win. A lottery based on the final standings can favor the worst teams only by giving a lower finish a better chance, so it always creates this problem. Equal odds avoids the problem by favoring no team. The NBA abandoned equal odds after five drafts because owners felt it did not help the worst teams [9]. The 3-2-1 lottery, which the league's owners approved 29 to 1 on May 28, 2026, starts with the 2027 draft and moves back toward equal odds [11]. It expands the lottery from fourteen teams to sixteen, with each team getting one, two, or three chances in the drawing. The drawing decides all sixteen picks, whereas the old lottery decided only the first four [7].

Most other proposals still judge teams on a single season and change which of its games count. Banchio and Munro would update the lottery odds after every game, then stop changing them when a team could first gain by losing [6]. We call that point the freeze. Applied to the 1988-89 season, their rule freezes the odds after the 332nd of 1,025 games and gives last-place Miami a 45.5% chance at the first pick. The odds then reflect how teams played before the freeze, regardless of what happened later. Charlotte finished with the second-worst record, yet it would have had a 7.5% chance. Indiana won eight more games than Charlotte and would have had an 18.5% chance, because more of Charlotte's losses came after the freeze. Kazachkov and Vardi [12] would rank the teams that miss the playoffs by where they stood at a point in the season fixed in advance. Lenten et al. [13] would give the first pick to the team eliminated earliest, and Gold [14] would give it to the team that wins the most games after it is eliminated. Banchio and Munro note that neither rule is fully incentive compatible [6].

A last group drops the lottery. The wheel, designed by Mike Zarren of the Boston Celtics, fixes each team's draft position years in advance. Over thirty years, each team picks first once, second once, and so on down to thirtieth [15], [16]. Losing buys nothing, and a weak team gets no extra help. Two other plans use an auction. Nate Silver [17] would replace the draft with player bidding. Each team would get between 25 and 100 units to bid with each year, and the worst records would get the most. A Boston Celtics proposal reported by Quinn [18] would have teams bid credits for draft positions. Unspent credits carry over to the next year, and a team loses 20 credits for each playoff series it wins.

None of the draft proposals above look beyond the current season to decide which teams need help. The Celtics' credits do, because they carry over and shrink when a team wins in the playoffs, but they work through an auction. COLA uses the same longer view to set the draft order directly, without an auction.

3. The COLA framework

3.1. How a COLA mechanism works

A COLA mechanism keeps one number for each team, called its priority, and updates that number once a year. After the playoffs, priority rises for the teams that went without success, which in most mechanisms means without a playoff series win. It falls for the teams that had success. The mechanism then sets the draft order from priority, and the teams that receive the best picks give some of their priority back. The next season starts where this one ended. A long run without success therefore builds priority up, and one good season or one high pick spends it.

Simple COLA shows this cycle in its plainest form. A team's priority is its wait, the number of seasons since it last won a playoff series or received one of the first three picks. A team whose last series win came six seasons ago, with no top-three pick since, has a wait of six. When the playoffs end, every team that won a series goes back to a wait of zero, and every other team adds one season. There are 22 such teams: the 14 that missed the playoffs and the 8 that lost in the first round. Those 22 teams pick first, in order of wait, longest first. If two teams have the same wait, the one with more wins this season picks first. The 8 teams that won a series pick after them. After the draft, the three teams that picked first go back to zero.

The mechanisms in the family differ in three choices. We call these choices dials, because a league can set each one separately. The first dial is how priority grows. In every mechanism we study, each team that qualifies gets the same fixed step, whatever its record. A team that already qualifies therefore gains nothing more by losing more games. What a team might gain by losing its way into qualifying is the subject of Section 5. The second dial is how priority falls, meaning how much a playoff run or a high pick costs. The third dial is how priority becomes lottery odds and draft positions. It covers which teams take part, a group we call the pool, and whether their order comes from a drawing, from priority alone, or from both.

3.2. Six COLA mechanisms

Simple COLA is the first of the six COLA mechanisms we test. Three others measure priority in lottery tickets, where a ticket is one chance in a drawing. A team gains 1,000 tickets in each season without a playoff series win. Each series it wins takes away a quarter of the tickets it held at the start of the playoffs. A team with 5,000 tickets that wins two series keeps 2,500, and a champion, with four series wins, keeps none. Picks cost tickets too. The first pick takes away all of a team's tickets, and the second, third, and fourth picks take away 75%, 50%, and 25%. In all three mechanisms, the pool is the 14 teams that missed the playoffs.

The three differ only in the third dial. Classic COLA draws the first four picks, giving each team a chance proportional to its tickets, and then hands out picks 5 to 14 in reverse order of record. Waitlist COLA draws the first four picks the same way and hands out picks 5 to 14 in ticket order, with the team that has the most tickets going first. Full-lottery COLA draws all 14 picks, one after another, with each team's chance proportional to its tickets.

The last two mechanisms reward both winning and waiting. Countdown COLA uses Simple COLA's wait and its pool of 22 teams. It scores each team by its wait multiplied by its wins this season, so among teams with the same wait, the one that won more scores higher. It then fills the draft order from the

last pick to the first. At each step, it draws one team from the five lowest scores not yet placed, and lower scores make a team more likely to be drawn. The teams with the best scores are placed last, so they pick first. Beckett COLA counts the seasons since a team last won a playoff series, finished in the top six of its conference, or received the first pick. Its pool is every team with a count of two or more. Each team gets entries equal to its count multiplied by its wins. The first four picks are drawn with chances proportional to entries, and the remaining pool picks in order of entries, most entries first.

3.3. Three reference lotteries

We compare the six COLA mechanisms with three lotteries that set priority from one season alone. The first is the lottery the NBA used from 2019 to 2026, which we call the old lottery. The second is an equal-odds lottery, in which all 14 picks among the teams that missed the playoffs are drawn with equal chances. The third is 3-2-1, as the NBA adopted it.

In the framework's terms, 3-2-1 rebuilds priority from this season's standings every year. Nothing grows with waiting, and success costs nothing in later years. The one thing it remembers is its limit on repeat picks. A team cannot receive the first pick two years in a row or a top-five pick three years in a row [7]. That places 3-2-1 at the edge of the family, with the first two dials set almost to zero. It also explains the two properties noted earlier. Its priority comes from one season's standings, so some teams can still gain by finishing lower. Nothing carries over, so a team that has been weak for years gets no extra help. Section 7 measures both.

Table 1 lists the dial settings of all nine mechanisms. Appendix B gives the exact rules, including how ties are broken and where the teams outside each pool pick.

Mechanism	Pool	How priority grows	How priority falls	How priority becomes picks
Old lottery	The 14 teams that missed the playoffs	Nothing carries over	Nothing carries over	First four picks drawn, with chances set by record. Picks 5 to 14 in reverse order of record.
Equal-odds lottery	The same 14 teams	Nothing carries over	Nothing carries over	All 14 picks drawn with equal chances.
3-2-1	16 teams: the 10 that miss the play-in, the 4 ninth and tenth seeds, and the 2 losers of the seven-eight games	Nothing carries over	Only the limit on repeat picks	All 16 picks drawn. The three worst records hold 2 chances each, the next seven hold 3, the ninth and tenth seeds 2, and the seven-eight losers 1.
Classic COLA	The 14 teams that missed the playoffs	1,000 tickets in each season without a playoff series win	A quarter of its tickets for each series won. Picks 1 to 4 cost 100%, 75%, 50% and 25%.	First four picks drawn by tickets. Picks 5 to 14 in reverse order of record.
Waitlist COLA	The same 14 teams	As Classic COLA	As Classic COLA	First four picks drawn by tickets. Picks 5 to 14 in order of tickets.
Full-lottery COLA	The same 14 teams	As Classic COLA	As Classic COLA	All 14 picks drawn by tickets.
Simple COLA	The 22 teams without a series win this season	One season in each season without a playoff series win	To zero after a series win or a top-three pick	No drawing. Longest wait first, and ties go to the team with more wins.
Countdown COLA	The same 22 teams	As Simple COLA	As Simple COLA	Score is wait times wins. The order is drawn from the last pick to the first, each time among the five lowest scores left.
Beckett COLA	Every team with a count of two or more	One in each season without a series win, a top-six finish in its conference or the first pick	To zero after any of those three	Entries are count times wins. First four picks drawn by entries, the rest in order of entries.

Table 1. The nine mechanisms: the pool of each and its three dial settings.

4. Incentive compatibility

Section 1 called a mechanism incentive compatible when no team can improve its pick by losing on purpose. For Simple COLA, we can state that property exactly, prove it, and say what the proof assumes.

4.1. The result

Under Simple COLA, a game can change the pool order in two ways. First, it can change which teams win a first-round playoff series. Those eight teams leave the pool and return to a wait of zero, so the list of series winners fixes which teams are in the pool and how long each has waited. The second is the tie-break, the rule that separates teams with the same wait. It uses each team's regular-season win total, and the pool order uses that total for nothing else. The theorem below addresses the second case, and Appendix A proves it.

Theorem 1. Take any season under Simple COLA and any team that finishes it in the pool. Suppose the team loses a regular-season game it could have won, and the same eight teams win a first-round series either way. Then the team's pick in that season's draft is the same as it would have been with the win, or later. It is never earlier.

Suppose two teams have each waited five seasons. One finishes with 30 wins, and the other with 28, so the 30-win team picks first of the two. Had it lost three more games, it would have finished with 27 wins and picked second. Its wait would still be five seasons, so it would still pick after every team that had waited six seasons or more and before every team that had waited four or fewer.

4.2. Why it holds

The proof has three steps. First, the loss changes no team's wait. A wait counts the seasons since a team last won a playoff series or received a top-three pick, and with the same series winners, every wait stays the same. Second, the loss changes two win totals. The team has one win fewer, and its opponent has one more. Third, the order compares wins only between teams with the same wait, and the team with more wins picks first. Every team that would have picked ahead of the losing team still does, and a team that was level with it or just behind may now pick ahead too. The number of teams picking before it can rise and cannot fall. A coin flip settles any tie on both wait and wins, so a loss can also turn a lead into a coin flip, or a coin flip into a place behind. The same steps apply one game at a time, so they cover any number of losses.

The impossibility result in Section 1 does not reach Simple COLA. That result covers lotteries that set odds based on final standings alone and give worse records odds at least as good as better ones. Simple COLA reads the standings in one place, the tie-break, and there it favors the team with more wins. Its help for weak teams comes from the wait, which rises by one season for every team in the pool, whatever its record. Simple COLA therefore does what the result rules out for a lottery based on the final standings. It favors the teams that have gone longest without success, and it gives no team in the pool an earlier pick for losing.

4.3. What the result assumes

Theorem 1 takes the list of series winners as given, and like the impossibility result, it looks at one season and its draft [6]. Three assumptions about how teams behave carry it to the claim that no team has a reason to lose a game.

1. A team prefers an earlier pick to a later one.
2. A team prefers winning a playoff series to going without one.

3. A team would rather win a game than lose it to change which other teams are in the pool.

The first two are standard. In Banchio and Munro's model, teams value the first pick and a place in the playoffs at least as much [6]. Section 5 takes up the cases in which either assumption could fail and how a league can guard against them.

The third assumption needs a defense, because a team's pick does depend on which other teams are in the pool. If a team that would have picked ahead leaves the pool, every team behind it moves up one place. One place is the most a team can gain from any one rival in this way.

Under Simple COLA, a rival leaves the pool only by winning a playoff series. Reaching the playoffs changes nothing in the draft, because a team that loses in the first round stays in the pool. A team that wanted a rival out would have to help it win a series, and the one kind of help that requires losing is a regular-season loss to that rival, which adds one win to the rival's total. In our simulated leagues, 14% of teams that won 40 to 44 games won a playoff series, and 41% of teams that won 45 to 49 games did. That is the steepest rise in the data, about five percentage points for each extra win. It overstates what one handed-over win is worth, because stronger teams both win more games and win more series. A deliberate loss therefore buys at most about one-twentieth of a draft place, and only when the rival is ahead in the order. The third assumption asks only that a team value a win more than that.

The loss also has costs inside the draft. It takes one win from the team's own total, and in the simulated leagues one more loss costs a team that missed the playoffs 0.15 places on average through the tie-break. The league also punishes such games, as the $500,000 fine in Section 1 shows [2].

4.4. A check on the simulated leagues

A proof covers the rule as written, so we also tested the rule as our simulator runs it. We first confirmed that the simulator's draft order followed the rule in all 750 seasons of the Simple COLA run: 25 seasons in each of 30 leagues. Then, in each season, we took every team that missed the playoffs except the one with the fewest wins. We lowered its win total one game at a time until it sat one win below that team, and we held everything else constant. That gave 158,325 cases. Counting each coin-flip tie as half a place, the team's place in the order was later in 98,695 cases and unchanged in the other 59,630. It was earlier in none.

5. When the assumptions fail

Section 4 called two of its assumptions standard. This section takes each in turn, shows where it could fail, measures the failure size in our simulated leagues, and describes what a league could do about it. No team in the simulated leagues loses on purpose, so each figure below measures a temptation and says nothing about how often a real team would act on it.

5.1. A team that prefers the pool to the playoffs

The second assumption says a team prefers winning a playoff series to going without one. It matters because every mechanism's pool ends at a line, and a team just inside the pool picks earlier than a team just outside. Table 2 shows that line under four mechanisms and what a place on the pool side is

worth.

Under the old lottery, the line is the last playoff place. The best team that missed the playoffs picks 13.6th on average, and the first pick outside the pool is the 15th, so staying in the pool is worth 1.4 places. Under 3-2-1, the line is the play-in game between the seventh and eighth teams in a conference. The loser gets one chance in the drawing and picks 11.6th on average; the first pick outside the pool is 17th, a difference of 5.4 places. Under both lotteries, the team on the line holds the worst odds in the pool.

The difference is larger under COLA, because priority does not depend on record. The team on the line can have the highest priority. Under Full-lottery COLA, the team with the most tickets among those that missed the playoffs picks 5.4th on average, 9.6 places ahead of the first pick outside the pool. Under Simple COLA, the line is a first-round series. A playoff team that holds the longest wait picks first if it loses that series and 23rd at best if it wins, a difference of 22 places. This is the second cost named in Section 1.

To collect that difference, a team would have to lose a playoff series on purpose and give up the second round. A league that prefers not to rely on teams refusing that trade has two options. The first is already in our tests. A full-lottery COLA ends its pool at the last playoff place and draws every pick, so the most a team can gain at the line is 9.6 places, and the gain comes with better odds in a drawing. The second is to count any playoff appearance as success. A first-round loss then buys nothing. That change has two costs. Help stops for a team that reaches the playoffs every year and never advances, and the line returns to the end of the regular season, where late losses can move a team across it. We have not simulated that variant.

Mechanism	Where the pool ends	Places gained by staying in the pool	Top-three picks that went to a team with one in the previous two drafts	Next pick after a top-three pick, when the team missed the playoffs again
Old lottery	Last playoff place	1.4	29%	6.6
3-2-1	Seven-eight play-in game	5.4	20%	8.6
Full-lottery COLA	Last playoff place	9.6	16%	9.4
Simple COLA	First-round series	22	0%	21.0

Table 2. Four mechanisms on this section's measures, from 30 simulated leagues of 25 seasons. For the two COLA mechanisms, the places gained are for the team with the highest priority, which is the most any team on the line can gain.

5.2. A team that prefers a later pick

The first assumption says a team prefers an earlier pick to a later one. Under Simple COLA, it can fail at one line in the order, between the third and fourth picks. A top-three pick returns a team's wait to zero, a step we call a reset, and the fourth pick leaves the wait untouched. When the teams on either side of that line have the same wait, only wins separate them, and the team with fewer wins picks fourth and keeps its wait.

Every league starts with no history, so all its teams share one wait at first. We therefore measured this line in seasons 6 to 25 of each league. The third and fourth teams shared a wait in 410 of 600 seasons,

or 68%. In the following season, the team that had picked third picked 21st on average, and the team that had picked fourth received a top-three pick 89% of the time. To change places with the fourth team, the third team would have needed a median of six more losses.

The exception is narrow. It concerns the teams tied at one line, about four on average, and it asks a team to trade the third pick now for the chance of a better one a year later. The simplest remedy is to break ties between teams with the same wait by a random draw. Wins then play no part in the order, so the exception disappears, at the price of a small drawing and of the tie-break's reward for winning. We have not simulated that variant. The ticket mechanisms avoid a single line in another way. They draw the first four picks and charge for them on a sliding scale, from all of a team's tickets for the first pick to a quarter for the fourth.

5.3. The season after a top pick

The reset behind that line is also what spreads the top picks among teams. Under the old lottery, 29% of top-three picks went to a team that had received one in the previous two drafts. The share was 20% under 3-2-1 and 16% under Full-lottery COLA. Under Simple COLA, it was zero.

The cost falls on a team that takes a top-three pick and stays weak. If it missed the playoffs again the next season, its pick that season averaged 21.0 under Simple COLA, against 6.6 under the old lottery, 8.6 under 3-2-1 and 9.4 under Full-lottery COLA. Such a team then waits. On average, 7.6 seasons pass between one team's top-three picks under Simple COLA, against 4.9 under the old lottery. Section 7 measures how this affects the time teams take to return to the playoffs. A league that wants a shorter wait after a top pick can soften the reset, as the ticket mechanisms do with their sliding scale.

5.4. Draft classes

No mechanism we test looks at how strong a draft class is. Under Simple COLA, a team's turn at the front comes when its wait says so, in a strong year or a weak one. A lottery leaves the same question to chance.

5.5. What a remedy costs

Each remedy in this section adds a rule to a mechanism whose appeal is that it has so few. Simple COLA keeps one count for each team and holds no drawing. A tie-break by random draw adds a small drawing. Full-lottery COLA adds tickets, percentages, and a drawing for every pick. The choice depends on how much weight a league gives to the exceptions above, and Sections 7 and 8 return to it by comparing all nine mechanisms.

6. Simulation design

A proof covers one season's draft. How much a mechanism helps the teams that wait, and what it does to long-term parity, shows only over many seasons, so we test the nine mechanisms in simulated leagues. This section describes the simulator, what we ran, what we measured, and how the simulated leagues compare with the NBA.

6.1. The simulator

ZenGM Basketball-GM is a basketball management game with public code [19]. Its user acts as a team's general manager, the person who decides which players the team drafts, signs, and trades.

The simulator creates a league of fictional players. Each player has ratings, numbers that describe a player's skills. The simulator plays every game one possession at a time, where a possession is one team's turn with the ball, and what happens on each possession depends on the ratings of the ten players on the court. It also runs everything around the games. Players improve, age, get injured, and retire. Teams sign players to contracts under a salary cap, a limit on the total a team pays its players. Teams also trade players and compete for free agents, the players whose contracts have ended. Each year, a new group of prospects, the young players available to be drafted, enters through a draft of two rounds, and each team starts with one pick in each round. We let the simulator's built-in general managers run all 30 teams, so no person makes a decision in any league.

A league has the shape of the NBA. Thirty teams in two conferences play 82 games each, followed by a play-in and playoffs for 16 teams, in which each series goes to the first team to win four games. Draft order matters in these leagues for the same reason it matters in the NBA. The simulated teams strongly favor the best prospects still available, and a team's results come from the players it has.

The simulator's developer added COLA in February 2026 as a draft format its users can choose [20]. In this study, our own code sets the first-round draft order under each of the nine mechanisms, and the simulator does everything else. The simulator has no rule that tells a team to lose, so no simulated team loses on purpose. Where we report a reward for losing, we compute what a team would have gained by losing, and no team in the leagues acts on it.

6.2. What we ran

We ran each of the nine mechanisms on 30 leagues for 25 seasons. That makes 6,750 simulated seasons and about 8.3 million regular-season games.

A simulated league depends on chance at every step, from the players it starts with to each shot and each lottery drawing. One number, called a random seed, fixes all of those chance events, so a given seed and mechanism always produce the same league. We used the same 30 seeds for every mechanism. A league therefore starts with the same players under all nine mechanisms and plays an identical first season. The nine versions of a league begin to separate at the first draft, and by the fourth season no two of them are alike.

Every league starts with no history, so every wait and every ticket count begins at zero. One approximation applies to 3-2-1. Its pool includes the loser of each seven-eight play-in game, and in our simulated leagues the team with the eighth-best record in each conference stands in for that loser. Appendix B gives the remaining details of each mechanism as we ran it.

6.3. What we measured

Section 1 listed three things we measure: help for the teams that have waited longest, long-term parity, and the reward for losing. This subsection makes them exact, with one number for the reward for losing and two each for help and parity.

Let p_i be the number of the pick that team i receives in a season's draft, so that $p_i = 1$ means the first pick. Under a mechanism with a drawing, p_i depends on chance, and we use its expected value $\mathrm{E}[p_i]$, the average pick over all the ways the drawing could come out. A team with an equal chance at the first pick and the third pick has an expected pick of 2. We compute each expected value exactly, by going through every possible outcome of the drawing and how likely it is, so the figure carries no random error.

The reward for losing. Take one season and the team with the fifth-worst record among the 14 that missed the playoffs. Call it team m. Suppose it had finished with the worst record in the league and nothing else had changed, so that every other team keeps its record and every team keeps its priority. Write $\mathrm{E}[p_m^{\text{worst}}]$ for its expected pick in that case. The reward for losing is

$$R = \mathrm{E}[p_m] - \mathrm{E}[p_m^{\text{worst}}].$$

If the team would pick sixth on average as things stand and fifth on average with the worst record, R is $+1$. A positive R means the worse record brought an earlier pick. A zero or negative R means losing bought nothing or cost the team places. The fifth-worst record is our standard starting point, and Section 7 reports the others.

The help for a longer wait. Take the team with the fifth-highest priority among the same 14 teams. Call it team q. Suppose its priority alone had been one step above the highest in its pool, where a step is one season of waiting or 1,000 tickets. Write $\mathrm{E}[p_q^{\text{top}}]$ for its expected pick in that case. The help for a longer wait is

$$H = \mathrm{E}[p_q] - \mathrm{E}[p_q^{\text{top}}].$$

It counts the draft places a mechanism gives a team for having waited the longest. Under a lottery in which nothing carries over, H is zero.

The longest wait. For each team, we count the seasons in a row in which it neither won a playoff series nor received a top-three pick. Write $w_i(s)$ for team i's count after season s and its draft, with $w_i(0) = 0$. Then

$$w_i(s) = \begin{cases} 0 & \text{if team } i \text{ won a series or received a top-three pick in season } s, \\ w_i(s-1) + 1 & \text{otherwise.} \end{cases}$$

This is Simple COLA's wait, counted the same way under every mechanism. The longest wait in a league is the largest count any team reaches in any season,

$$L = \max_{i,\,s} w_i(s).$$

Two waits for playoff success. A stretch of seasons without a series win ends when the team wins a series. A team that wins a series, goes four seasons without one, and then wins again has completed a stretch of four seasons. The wait for the next series win is the average length of the stretches that ended during a league's 25 seasons. The wait to return to the playoffs is the same average for stretches of seasons outside the playoffs. A stretch still open when the league ends is left out, under every mechanism alike. Shorter waits mean that a team that falls behind recovers sooner, so these two numbers measure long-term parity.

Each measure gives one number per league. R and H are averaged over the league's 25 seasons, and the other three are computed once per league. We report the average over the 30 leagues.

6.4. How we compare two mechanisms

To compare mechanisms A and B on a measure X, we take the difference in each league, $d_\ell = X_\ell^A - X_\ell^B$ for leagues $\ell = 1, \ldots, 30$. Comparing two mechanisms league by league in this way is called pairing. We report the average difference $\bar{d}$ with a 95% confidence interval,

$$\bar{d} \pm t \frac{s_d}{\sqrt{30}},$$

where s_d is the standard deviation of the 30 differences, a measure of how far apart they lie, and $t = 2.045$ is the standard multiplier for a 95% interval built from 30 values (the 97.5th percentile of Student's t distribution with 29 degrees of freedom). When the interval excludes zero, we say the two mechanisms differ on that measure.

Each league counts once, however many seasons it holds, because the seasons of one league depend on each other. Pairing guarantees that both mechanisms start from the same 30 leagues, and its effect fades after the first draft. Across the 30 leagues, a league's longest wait under one mechanism is almost unrelated to its longest wait under another. The correlation is a number between -1 and 1 that measures how closely two quantities move together, with 0 meaning they are unrelated. For the longest wait, it averages -0.02 over the 36 pairs that nine mechanisms make. For the two average waits, it averages 0.01 and -0.01. The intervals are therefore about as wide as intervals from unrelated leagues would be.

6.5. How the simulated leagues compare with the NBA

The leagues are fictional, so we checked them against the NBA in two ways. The NBA figures come from the standings, playoff results, and draft picks on Basketball Reference, a public archive of NBA statistics [21].

The first check compares league-wide patterns. Table 3 sets the 30 leagues we ran under the old lottery beside the NBA's last 25 seasons, 2001-02 to 2025-26. We use those leagues because the NBA held a lottery weighted by record in every one of those seasons. A team's win percentage is the share of its games that it won.

Statistic	NBA	Simulated leagues
Spread of win percentage within a season (standard deviation)	0.152	0.135 (0.121 to 0.153)
Correlation between a team's win percentage and its own a season earlier	0.60	0.62 (0.47 to 0.70)
The same correlation, three seasons apart	0.09	0.29 (0.17 to 0.40)
Playoff teams that reached the playoffs again the next season	73%	73% (67% to 78%)
Wait to return to the playoffs, in seasons	2.99	2.89 (2.25 to 3.36)
Wait for the next series win, in seasons	4.55	4.65 (3.93 to 5.93)
Longest wait for a series win or a top-three pick, in seasons	13	16.4 (12 to 25)

Table 3. The 30 leagues run under the old lottery beside the NBA's 25 seasons from 2001-02 to 2025-26. The last column gives the average of the 30 leagues, with the lowest and highest league in parentheses.

The simulated leagues match the NBA on four of the seven statistics. Playoff teams return at the same rate. The two average waits are within about a tenth of a season of the NBA's. A team's record also follows its record of a season earlier as closely as in the NBA.

They differ in two ways. Records are a little more bunched in the simulated leagues, where the spread is 0.135 against the NBA's 0.152. Some NBA teams lost on purpose in those seasons and no simulated team does, which may explain part of that gap. Team strength also lasts longer there. Three seasons apart, the correlation is 0.29, against 0.09 in the NBA, a value lower than in any of the 30 leagues. A weak simulated team therefore stays weak for longer, and the longest waits run longer. Under the old lottery, the longest wait averages 16.4 seasons across the simulated leagues. The NBA's longest over the same 25 seasons was 13, by Sacramento from 2004-05 to 2016-17.

Section 7 should be read with this in mind. Every mechanism is tested on the same leagues, so its comparisons between mechanisms are on equal terms. The lengths of the longest waits under the three reference lotteries are likely somewhat greater than the NBA would see. Section 8 returns to what one simulator can show.

The second check asks whether the reward for losing and the help for a longer wait come out the same on real standings. We took each of the 22 seasons from 2004-05, the first with 30 teams, to 2025-26. For each one, we computed R and H from that season's final standings and playoff results alone, with the same code we use on the simulated leagues. Each team's wait and its earlier picks come from the real seasons since 1999-00. We do not replay history. No mechanism's picks are allowed to change what happened in later seasons, so every real result stays as it happened. The check asks what each mechanism's rule would have done with the standings the NBA actually produced. We apply the old lottery's odds, which date from 2019, to all 22 seasons.

Measure	**Mechanism**	**Real standings: average (95% interval)**	**Simulated leagues**
Reward for losing, R	Old lottery	$+1.30$ (1.26 to 1.34)	$+1.32$
	3-2-1	-0.55 (-0.70 to -0.39)	-0.63
	Simple COLA	-0.73 (-1.07 to -0.39)	-0.94
Help for a longer wait, H	Simple COLA	6.0 (5.4 to 6.7)	6.3

Table 4. The reward for losing and the help for a longer wait on the NBA's real standings, one season at a time, beside their values in the simulated leagues. All values are draft places. Each interval treats the 22 seasons as unrelated to one another.

Table 4 shows that all four simulated values lie inside the intervals from the real standings. In all 22 real seasons, Simple COLA's reward for losing was zero or negative, as Theorem 1 requires, and the old lottery's was positive.

Simple COLA's two values depend on waits, and a simulated league starts with every wait at zero. From the sixth season of each league on, the simulated teams have the same five seasons of history behind them as our data holds for the real teams in 2004-05. Measured over those seasons, the two values are -0.57 and 6.0, and both are inside the intervals too.

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